Lecture 11

Tuesday, 14 February 2023 1:29 PM

## Phase + Reaction Equilibrium for Electrochemical systems

$$\underset{\alpha}{Pt} \,|\, \underset{\beta}{H_2(g)} \,|\, \underset{\gamma_1}{H_2SO_4(aq)} \,||\, \underset{\gamma_2}{CuSO_4(aq)} \,|\, \underset{\delta}{Cu(s)} \,|\, \underset{\alpha'}{Pt(s)}$$

→ When the potential difference b/w RHS + LHS electrode ($V^{\alpha'} - V^{\alpha}$) creates equilibrium, we define this as:

$$U_{cell} \equiv V^{\alpha'} - V^{\alpha}$$

→ We also know that measured potential differences across electrodes describe the <u>electrochemical potential of $e^-$</u>

$$\mu_e^{\alpha} - \mu_e^{\alpha'} = \frac{-F(V^{\alpha} - V^{\alpha'})}{n} \quad [=] \frac{J}{mol\, e^-} \quad ; \text{ Thus } \boxed{\mu_e^{\alpha} - \mu_e^{\alpha'} = FU} \text{ @ equilibrium}$$

Left − Right

$$W_{rev} = \Delta G_r = -F\Delta V$$

→ Applying phase equilibrium to our examples means that species in adjacent phases must be in equilibrium:

So on RHS: $\underset{\delta}{Cu(s)} \,|\, \underset{\alpha'}{Pt(s)}$

$$\mu_e^{\delta} = \mu_e^{\alpha'} \text{ @ equil}$$

Rxn: $\underset{\delta}{Cu^\circ} \rightleftharpoons \underset{\gamma_2}{Cu^{2+}} + \underset{\delta}{2e^-}$

$$2\mu_{e^-}^{\delta} = \mu_{Cu}^{\delta} - \mu_{Cu^{2+}}^{\gamma_2}$$

$$\therefore 2\mu_e^{\alpha'} = 2\mu_e^{\delta} = \mu_{Cu^\circ} - \mu_{Cu^{2+}}^{\gamma_2}$$

→ Repeating this process for LHS (now only $e^-$ conducting phase $\alpha$)

Rxn: $H_2(g) \rightleftharpoons 2H^+_{(aq)} + 2e^-_{\alpha}$

$$2\mu_e^{\alpha} = \mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma_1}$$

→ Combining w/ cell eq. potential:

$$U_{cell} = \frac{1}{F}(\mu_e^{\alpha} - \mu^{\alpha'}) = \frac{1}{2F}(2\mu_e^{\alpha} - 2\mu_e^{\alpha'})$$

$$\boxed{U_{cell} = \frac{1}{F}\left(\mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma_1} - \mu_{Cu}^{\delta} + \mu_{Cu^{2+}}^{\gamma_2}\right)}$$

Don't typically measure $\mu$ → apply def. of activities to get a usable form:

$$\mu_{H_2}^{\beta} = \mu_{H_2}^{*} + RT \ln P_{H_2}^{\beta}$$

($\mu^*$: standard chem potential for a gas)

$\mu^{\delta}$ . . . $\mu^{\circ}$

$\mu_{H_2}^{\beta} = \mu_{H_2}^{\circ} + RT \ln P_{H_2}^{\beta}$ for a gas

$\mu_{Cu}^{S} = \mu_{Cu}^{\circ}$

$\mu_{H^+}^{\gamma_1} = RT \ln(\lambda_{H^+}^{\gamma}) = RT \ln(c_{H^+}^{\gamma_1} f_{H^+}^{\gamma_1} a_{H^+}^{\circ})$

$\mu_{Cu^{2+}}^{\gamma_2} = RT \ln(c_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}}^{\gamma_2} a_{Cu^{2+}}^{\circ})$ ⟶ Independent of conc.

$$V_{cell} = \frac{1}{2F}\left[\mu_{H_2}^{\beta} + RT \ln(P_{H_2}^{\beta}) - 2(RT \ln(a_{H^+}^{\circ})) + RT \ln(C_{H^+} f_{H^+}) - \mu_{Cu}^{\circ} + RT \ln(a_{Cu}^{\circ}) + RT \ln(C_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}})\right]$$

$V_{cell}^{\circ}$ → standard equilibrium potential ⇒

$$V_{cell}^{\circ} = \frac{1}{2F}\left(\mu_{H_2}^{\circ} - 2RT \ln(a_{H^+}^{\circ}) - \mu_{Cu}^{\circ} + RT \ln(a_{Cu^{2+}}^{\circ})\right)$$

$$= \underbrace{-\frac{1}{2F}\left(\mu_{Cu}^{\circ} - RT \ln(a_{Cu^{2+}}^{\circ})\right)}_{V_{RHS}^{\circ} \text{ (red)}} - \underbrace{\left(-\frac{1}{2F}\left(\mu_{H_2}^{\circ} - 2RT \ln a_{H^+}^{\circ}\right)\right)}_{V_{LHS}^{\circ} \text{ (ox)}}$$

Factor out minus signs so we can write $V_{cell}^{\circ} = V_{red}^{\circ} - V_{ox}^{\circ}$ s.t. $V > 0$

$$\boxed{V_{cell}^{\circ} = V_{RHS}^{\circ} - V_{LHS}^{\circ}}$$

Standard potentials, when all species are @ standard state: 1 M for solutes, 1 bar for gases.

$$\therefore V_{cell} = V_{cell}^{\circ} + \frac{RT}{2F}\left(\ln(P_{H_2}^{\beta}) - 2\ln(C_{H^+}^{\gamma_1} f_{H^+}^{\gamma_1}) + \ln(C_{Cu^{2+}}^{\gamma_2} f_{Cu^{2+}}^{\gamma_2})\right)$$

* assume all $f_j = 1$:

$$\boxed{V_{cell} = V_{cell}^{\circ} + \underbrace{\frac{RT}{2F} \ln\left(\frac{P_{H_2}^{\beta} C_{Cu^{2+}}^{\gamma_2}}{(C_{H^+})^2}\right)}_{\text{Nernstian correction}}}$$

$\eta = V - U$

This was a half cell potential (for an electrode) → Not the same as $V_{cell}$

To discuss half cell potentials we reference to a standard Hydrogen Electrode (SHE) $[V_{H_2|H^+}^{\circ}]$

an ideal solution @ equil exposed to 1 bar $H_2$ & 1 M $H^+$ (aq) [pH = 0]

Cx: $AgCl + e^- \rightleftharpoons Ag + Cl^-$ in ref to SHE

$\alpha \quad \beta \quad \gamma \quad S \quad \beta' \quad \alpha'$

$Pt \mid H_2(g) \mid HCl(aq) \mid AgCl(s) \mid Ag(s) \mid Pt(s)$

Phase eq: $\mu_{e^-}^{\alpha'} = \mu_{e}^{\beta'} = \mu_{e}^{S}$

$AgCl + e^- \rightleftharpoons Ag + Cl^-$ ⟶ This would be an equivalent reaction to look at (thermodynamic analysis will lead to an answer both ways)

Phase eq: $\mu_{e^-} = \mu_e - \mu_e$

$AgCl + e^- \rightleftharpoons Ag + Cl^-$ → This won't be equivalent reaction to keep as (thermodynamic analysis will lead to answer both ways)

RHS Rxn: $Ag \rightleftharpoons Ag^+ + e^-$ → $\mu_{e^-}$

$\mu_{Ag}^{\beta'} = \mu_{Ag^+}^{\delta} + \mu_{e^-}^{\alpha'}$

LHS Rxn: $H_2 \rightleftharpoons 2H^+ + 2e^-$

$\mu_{H_2}^{\beta} = 2\mu_{H^+}^{\gamma} + 2\mu_e^{\alpha}$

$$\mu_e^{\alpha} - \mu_e^{\alpha'} = F V_{cell}$$

$$\Rightarrow \left(\mu_{H_2}^{\beta} - 2\mu_{H^+}^{\gamma}\right) - \left(\mu_{Ag}^{\beta'} - \mu_{Ag^+}^{\delta}\right) = F V_{cell}$$

$$V_{cell} = \frac{1}{2F}\left(2\mu_e^{\alpha} - \mu_e^{\alpha'}\right)$$

Half-reaction values

Note 1: If we tabulate all rxns vs. SHE, then we can determine cell potentials + Nernst correction for any comb. of half rxns

$$V^\circ_{cell} = V^\circ_{Ag/Ag} - V^\circ_{Cu/Cu^{2+}}$$

$$\boxed{V^\circ_{cell} = \left(V^\circ_{Ag/Ag} - V^\circ_{H_2/H^+}\right) - \left(V^\circ_{Cu/Cu^{2+}} - V^\circ_{Cu/Cu^{2+}}\right)}$$

tabulated — tabulated → Standard potentials (or standard reduction potential)

↓ ↓

There are half-reactions

Note 2: If we experimentally measure electrode rxns @ standard conditions then we can just use the Nernst correction for deviations from tabulated standards.

ex: $Cu \rightleftharpoons Cu^{2+} + 2e^-$ in 0.0001M HCl in cell w/ SHE

$$V_{cell} = V^\circ_{cell} + Nernst_{cell} = V^\circ_{cell} + \frac{RT}{2F}\ln Cu^{2+}$$ ★

($V^\circ_{cell}$ = $(V^\circ_{Cu/Cu^{2+}} - V_{SHE})$ ↳ set to zero)

★ } If Pt is electrode (then electrolyte conc. matters) → That is why we correct for that

The tabulated values are particularly useful when conducting experiments, bc it's often inconvinient to experimentally set up SHE

↳ Indeed we can set up a convinient electrode like Ag/AgCl in KCl & ref our WE to that:

$$(V^\circ_{Ag/AgCl} - V^\circ_{H_2/H}) = 0.222\ V$$

or, $V^\circ_{Ag/AgCl} = 0.222V$ vs. SHE ||

Eg: Ref. will have liquid junction potential (across the membrane)

Ag/AgCl Ref: $Cu \rightleftharpoons Cu^{2+} + 2e^-$

↑ want this to be in eq.

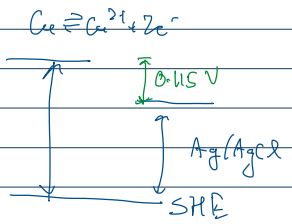
$Cu \rightleftharpoons Cu^{2+} + 2e^-$ → Finicky

Ag/AgCl Ref ; $Cu \rightleftharpoons Cu^{2+} + 2e^-$

↑ want this to be in eq.

0.222 V

$U^\circ_{Cu/Cu^{2+}} = 0.337$




$V_{Cell} = 0.337 - 0.222\ V = 0.115\ V$

A potentiostat provides ΔV b/w WE and Reference electrode → measures the I and then provides a ΔV at CE to match that i.

Tafel kinetics:

$$i_a = i_0 \exp\left(\frac{\alpha F \eta}{RT}\right)$$

$$\frac{15 \times 10^{-3}}{1 \times 10^{-4}} = \exp\left(\frac{(1.5)96485\,\eta}{RT}\right)$$

$$\frac{\ln(15 \times 10)\,RT}{(1.5)(96485)} = \eta$$

↑ vs. REF

0.115

$V_{SHE} = \eta + 0.222\ V$

0.115

$\eta = V - U$

$\eta$